



Allogeneic umbilical cord blood-derived mesenchymal stem cells combined with high tibial osteotomy: a retrospective study on safety and early results

Young-Woo Chung¹ · Hong-Yeol Yang² · Sung-Ju Kang² · Eun-Kyoo Song² · Jong-Keun Seon²

Received: 7 May 2020 / Accepted: 9 October 2020
© SICOT aisbl 2020

Abstract

Background Cartilage repair performed as a single-stage procedure is an important advancement in the treatment of full-thickness cartilage injury and has potential for widespread clinical use.

Purpose To investigate the short-term outcomes and cartilage regeneration after implantation of allogeneic human umbilical cord blood-derived mesenchymal stem cells (hUCB-MSCs) in patients who received high tibial osteotomy (HTO) for symptomatic medial knee osteoarthritis.

Methods Patients underwent treatment of full-thickness chondral injury in the osteoarthritic knee with HTO and implantation of hUCB-MSCs and were followed prospectively for a minimum of one year. Ninety-three patients were followed for a mean 1.7 years (range, 1.0–3.5). Median cartilage lesion size was 6.5 cm² (range, 2.0–12.8). Clinical outcomes were examined with patient-reported scoring instruments that consisted of the International Knee Documentation Committee (IKDC) subjective score, Western Ontario and McMaster University Osteoarthritis Index (WOMAC) score, Knee Society Score (KSS), and Hospital for Special Surgery (HSS) score. Cartilage regeneration was evaluated using the International Cartilage Repair Society (ICRS) cartilage repair assessment grading (CRA) system and the Koshino regeneration staging system in 49 patients who underwent second look arthroscopic assessment when their HTO plates were removed.

Results At final follow-up, the median IKDC subjective score had significantly improved from 39.0 to 71.3; the WOMAC score from 44.5 to 11.0; the KSS pain and function scores from 29.8 to 43.2 and 61.0 to 81.2, respectively; and the HSS from 61.6 to 82.7 ($p < 0.05$). Pre-operative examination showed ICRS grade IV cartilage injury in all knees, and cartilage regeneration at 2nd look arthroscopy showed improvements (8.2% of patients improved to ICRS grade I, 69.3% to grade II, and 22.5% to grade III). Moreover, Koshino stage was B in 24.5% and C in 75.5% of patients ($p < 0.05$).

Conclusion Allogeneic hUCB-MSC implantation combined with HTO for medial knee osteoarthritis was safe and showed signs of cartilage status improvement. Furthermore, randomized controlled studies with a control group are necessary to determine the real effectiveness and indications of this new combined procedure for patients with osteoarthritis.

Keywords Knee · Articular cartilage · Umbilical cord blood · Stem cell therapy · Osteotomy

Young-Woo Chung and Hong-Yeol Yang contributed equally to this work.

Level of evidence: 4

✉ Jong-Keun Seon
seonbell@jnu.ac.kr

¹ Department of Orthopedic Surgery, Gwangju Veterans Hospital, Gwangju, Republic of Korea

² Center for Joint Disease, Chonnam National University Bitgoeul Hospital, 80, Deoknamgil, Nam-gu, Gwangju, Republic of Korea

Introduction

High tibial osteotomy (HTO) is a good treatment option for moderate isolated medial unicompartmental osteoarthritis (OA) [1]. In general, HTO may be suitable for treatment of young, active patients [2], and long-term survival and functional outcomes after HTO are good to excellent [3, 4]. As a result of improvement in the mechanical environment within the knee, biological remodeling including regeneration of articular cartilage is induced after HTO [5, 6]. However, because the effects of cartilage repair in patients undergoing HTO are controversial, several concomitant techniques, such

as microfracture surgery and bone marrow aspirate concentrate (BMAC) application, have been used to obtain optimal cartilage regeneration. A recent study reported that the quality of cartilage repair after microfracture with collagen augmentation was superior to that after microfracture alone in patients undergoing HTO [7]. Hence, cartilage repair procedures performed concurrently with HTO might improve cartilage repair as observed in arthroscopic and histologic findings without increasing the risk of operative complications [8].

The use of stem cells for cartilage repair has become a major interest. Many human tissues, including bone marrow, adipose tissue, umbilical cord blood, and synovium, are well-known sources of adult mesenchymal stem cells (MSCs) [9]. Although there have been some reports of stem cell therapy applied in osteoarthritis, they have predominantly been the result of medical procedures and have described the application of heterogeneous cell populations obtained from autologous tissue containing uncertain portions or numbers of stem cells [10, 11]. Recently, human allogeneic umbilical cord blood-derived MSCs (hUCB-MSCs) have become commercially available. Although their use is thought promising, there is little clinical evidence of their effectiveness when implanted to treat chondral defects of the knee in OA.

Hence, the purpose of the present study was to investigate clinical outcomes after implantation of hUCB-MSCs in patients undergoing HTO for symptomatic medial knee OA. This study also evaluated cartilage regeneration status using second-look arthroscopy and compared clinical outcomes based on cartilage status. It was hypothesized that implantation of hUCB-MSCs would lead to excellent clinical outcomes and good cartilage regeneration.

Materials and methods

This retrospective case series study was conducted at one hospital between January 2016 and March 2019 after obtaining Institutional Review Board and Ethics Committee approval (CNUHH-2020-087). The data were prospectively collected from patients who received implantation of allogeneic hUCB-MSCs and hyaluronic acid (HA) hydrogel (Cartistem; Medipost Co., Seongnam-si, Korea) in addition to HTO. The hUCB-MSCs were isolated and characterized according to previously published methods [12].

The present study included patients younger than 65 years old with International Cartilage Repair Society (ICRS) grade III or IV cartilage defects ($> 2 \text{ cm}^2$), mechanical femorotibial varus angles $> 3^\circ$, and Kellgren-Lawrence (KL) grade 3 osteoarthritis. Those patients with severe obesity ($\text{BMI} \geq 35$); with histories of previous ipsilateral knee fracture; who had received other cartilage repair procedures, including microfracture or BMAC; who had received combined ligament surgery; with cartilage lesions affecting the lateral

compartment or patellofemoral compartments ($> \text{ICRS grade II}$); and who were followed up for less than 12 months were excluded. Initially, 99 patients were included. During the follow-up, four patients could not be contacted, and two died. The cohort ultimately included 93 patients. Patients completed initial and follow-up surveys. Surveys included the International Knee Documentation Committee (IKDC) subjective score, Western Ontario and McMaster University Osteoarthritis Index (WOMAC) score and pain/function of Knee Society Score (KSS) pain and function scores, and Hospital for Special Surgery (HSS) score. Follow-up responses were compared with baseline in all patients. Ninety-three patients were followed prospectively for a mean 1.7 years (range, 1.0–3.5). The mean age of the participants was 56.6 years (range, 43–65), and their mean BMI was 25.8 kg/m^2 (range, 20.9–33.2). The average defect size was 6.5 cm^2 (range, 2.0–12.8).

Preparation of MSCs

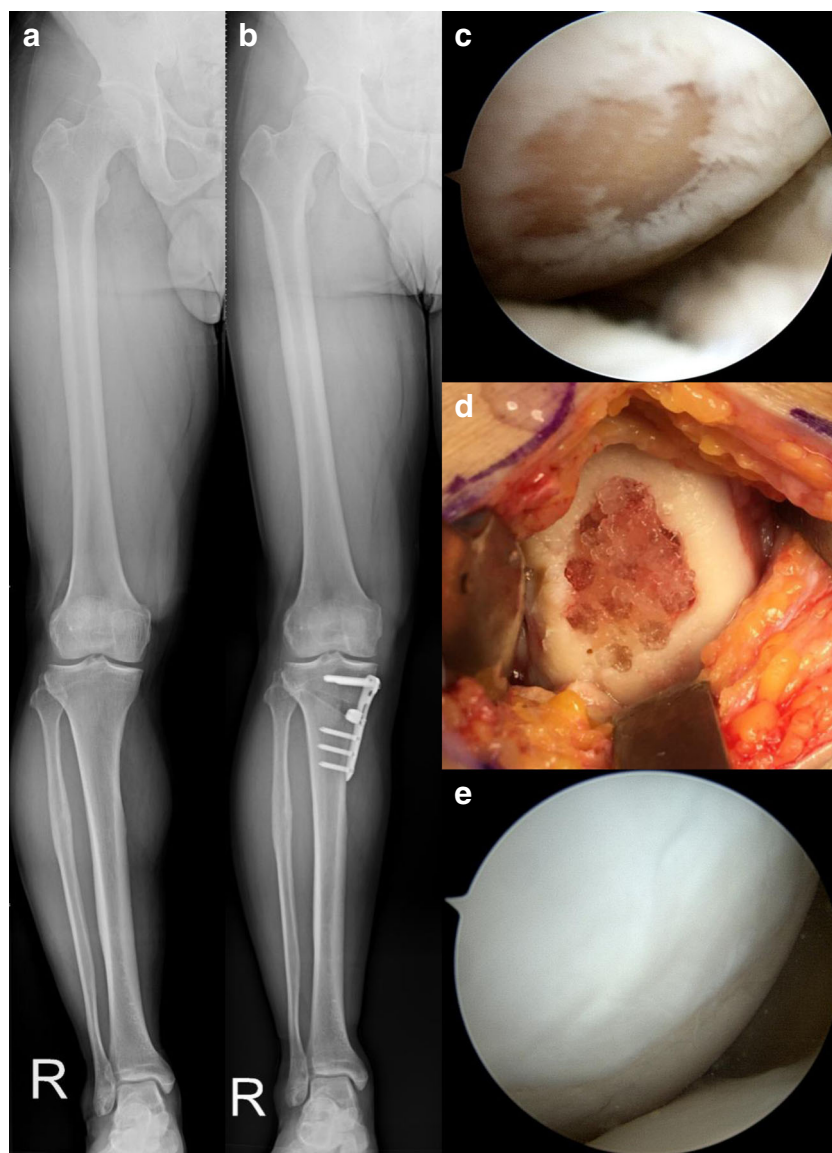
hUCB-MSCs A medicinal product (a composite of hUCB-MSCs and HA hydrogel) was produced under regulatory authority, approved by Good Manufacturing Practice guidelines, and provided as a new investigational drug by Medipost. Human umbilical cord blood was collected from umbilical veins at the time of neonatal delivery, with informed consent from the mothers, and stored in a cord blood bank. Mononuclear cells were separated by density gradient centrifugation at $550g$ for 30 minutes using a Ficoll-Hypaque solution. Isolated mononuclear cells were washed, suspended in minimum essential medium supplemented with 10% fetal bovine serum, and then seeded into culture flasks at a concentration of $53,106 \text{ cells/cm}^2$. Cultures were maintained at 37°C in a humidified atmosphere containing 5% CO_2 , with a change of culture medium twice per week. Colonies of spindle-shaped cells formed approximately two weeks after plating. Cells were trypsinized, washed, and resuspended in culture medium when the monolayer of MSC colonies reached 80% confluence. Therapeutic use of Cartistem for cartilage repair was reviewed and approved by the Korea Food and Drug Administration in January 2012 [13]. This product consists of 1.5 ml of cord BM-MSCs (7.5×10^6) and 4% HA. hUCB was collected from maternal umbilical veins during delivery, with informed consent from pregnant mothers and stored at a cord blood bank. Then, via repeated subcultures, hUCB-MSCs were isolated from hUCB and expanded [14]. hUCB-MSCs and HA were mixed according to the manufacturer instructions prior to application during surgery. hUCB-MSCs were administered at a therapeutic dosage of 500 mL/cm^2 of defect with a cell concentration of $0.5 \times 10^7 \text{ cells/mL}$.

Surgical techniques and postoperative management

A standard arthroscopic examination was performed to assess cartilage defects according to the ICRS grade system. When the surgeon judged them necessary, arthroscopic procedures, including meniscectomy or meniscus repair, were performed to treat meniscal problems. Biplanar opening wedge osteotomy was then performed according to established procedures using a locking wedge plate [15, 16]. The Fujisawa point was the valgus target point of HTO and was checked by intra-operative fluoroscopy [17]. After HTO, the medial femoral condyle (MFC) was exposed by mini-arthrotomy. Cartilage defects were prepared by debridement down to the beds using curettes until healthy-looking underlying bone was visible. Subsequently, multiple drill holes 4 mm deep and

4 mm in diameter were created approximately 2–3 mm apart using a drill bit (Medipost) in cartilage defect sites at the MFC. Between these holes, smaller holes 2 mm in diameter were created. The hUCB-MSC and HA hydrogel composite was then implanted into the holes in the cartilage defects from the base to the surface [13, 18] (Fig 1). The rehabilitation protocol started one day after the operation. With the help of knee braces, quadriceps strengthening exercises were allowed. Gradually, knee flexion from 0 to 90° was allowed after two post-operative weeks, while non-weight-bearing to partial weight-bearing walking with crutches was advised after six weeks isometric knee. Full weight-bearing was recommended based on the level of bone healing according to radiographic findings after 12 weeks.

Fig. 1 A 52-year-old man received HTO followed by second-look arthroscopy for plate removal after at least 1 post-operative year. **a** Pre-operative standing whole-lower-extremity radiograph with varus limb alignment. **b** Post-operative standing whole-lower-extremity radiograph showing correction of varus deformity and valgus limb alignment after HTO. **c** Arthroscopic image taken during the osteotomy showing a cartilage defect of the medial femoral condyle. **d** After debridement of the cartilage defect, the hUCB-MSCs-HA hydrogel composite was implanted into holes drilled into the cartilage defect from the base to the surface. **e** Second-look arthroscopic images taken 1.4 years postoperatively at the time of plate removal showing good coverage of the defect with smooth cartilage and complete integration (ICRS grade I)



Cartilage regeneration assessment

After confirming bony union at the osteotomy sites with anterior-posterior (AP) radiographs a minimum of 1 year after index surgery, second-look arthroscopic examination was performed simultaneously with removal of plates. Cartilage repair status was assessed under arthroscopy according to the ICRS-Cartilage Repair Assessment (CRA) and Koshino staging system [6, 19] (Fig 1). The CRA evaluates three items, namely, degree of defect repair, integration to border zone, and macroscopic appearance, and is graded as follows: normal as grade I, nearly normal as grade II, abnormal as grade III, and severely abnormal as grade IV. Arthroscopic findings were evaluated by two independent observers who did not directly participate in the surgical procedures and were blinded to the study aim. Regenerated articular cartilage was classified into the good regeneration group, with ICRS grade I or II (normal or nearly normal), or the poor regeneration group, with ICRS grade III or IV (abnormal or severely abnormal).

In addition, the status of the regenerated articular cartilage was also evaluated according to the macroscopic staging system described by Koshino et al. [6]: stage A, no regenerative change; stage B, pink fibrous tissue with or without partial coverage and with white fibrocartilage; and stage C, cartilage with total regeneration. The grade was assessed on the basis of a consensus between trained orthopedic fellows who did not directly participate in the surgical procedures and were blinded to the study aim (ICC, 0.75–0.87).

Statistical analysis

Analysis was performed using IBM SPSS Statistics for Windows (version 22; IBM, Armonk, NY, USA). The Kolmogorov-Smirnov test was used to check whether scores were normally distributed or not. The pre- and post-operation survey scores were compared using the paired *t* test for normally distributed data. For intergroup comparison of clinical outcomes based on cartilage regeneration, the Mann-Whitney test was used for non-normally distributed data. *p* values less than 0.05 were considered statistically significant.

Results

At final follow-up, the median IKDC subjective score had significantly improved from 39.0 to 71.3; the WOMAC score from 44.5 to 11.0; the KSS pain and function scores from 29.8 to 43.2 and 61.0 to 81.2, respectively; and the HSS from 61.6 to 82.7 (*p* < 0.05). Table 1 summarizes the clinical outcomes of hUCB-MSCs combined with HTO between pre-operative and final follow-up assessment.

Table 1 Clinical outcomes of HTO with hUCB-MSC application

	Pre-operative	Final follow-up	<i>p</i> value [†]
IKDC	39.0 ± 10.4 (12–73)	71.3 ± 5.9 (58–86)	< 0.001
WOMAC			
Pain	9.5 ± 3.8 (2–20)	2.6 ± 1.5 (0–6)	< 0.001
Stiffness	4.2 ± 1.6 (2–9)	2.2 ± 1.5 (0–8)	< 0.001
Function	33.8 ± 10.9 (12–62)	6.0 ± 2.4 (1–13)	< 0.001
Total	44.5 ± 15.1 (19–87)	11.0 ± 3.7 (1–27)	< 0.001
KSS			
Pain	29.8 ± 11.8 (0–50)	43.2 ± 5.0 (20–50)	< 0.001
Function	61.0 ± 16.3 (0–100)	81.2 ± 13.7 (50–100)	< 0.001
HSS	61.6 ± 12.9 (27–98)	82.7 ± 13.5 (31–100)	< 0.001

*Values are given as means and standard deviations, and a *p* value of < 0.05 was considered significant

[†] Paired *t* test for analysis of difference

IKDC International Knee Documentation Committee, WOMAC Western Ontario and McMaster Universities Osteoarthritis Index, KSS Knee Society Score, HSS Hospital for Special Surgery

Second-look arthroscopic cartilage regeneration findings

At an average of two years after initial surgery, 49 patients underwent second-look arthroscopy with plate removal. According to the ICRS-CRA grading system, the mean total score was 9.3 points (range 4–12) (Tables 2). Second-look arthroscopy demonstrated ICRS grade I in four (8.2%) patients, grade II in 34 (69.3%), and grade III in 11 (22.5%) (Table 2). In terms of Koshino staging, 12 (24.5%) patients showed partial coverage, 27 (55.1%) full coverage with overgrowth, and 10 (20.4%) full coverage with even growth (Table 2).

Comparison of clinical outcomes based on cartilage regeneration status

As shown in Tables 3, there were no significant differences in clinical outcomes (IKDC, WOMAC score, KSS pain and function scores, and HSS) between groups based on cartilage regeneration status according to either the ICRS grading system or the Koshino staging system.

Complications

No persistent effusion, synovitis, localized eruption, or localized erythema that was considered an allergic reaction after hUCB-MSC implantation was noted. Some patients exhibited knee swelling for up to 1 month after surgery and required analgesic medication, but these episodes were self-limiting. There was no infection, neurovascular injury, deep vein thrombosis, or non-union associated with HTO.

Table 2 Second-look arthroscopic findings after HTO with hUCB-MSC application

Stage of regeneration	N (%)
ICRS Grading System [‡]	
Grade 1: Normal	4 (8.2)
Grade 2: Nearly normal	34 (69.3)
Grade 3: Abnormal	11 (22.5)
Grade 4: Severely abnormal	
Koshino staging system	
Stage A: No regeneration or no repair	
Stage B: Pink fibrous tissue with or without partial coverage with white fibrocartilage	12 (24.5)
Stage C1: Total coverage with white overgrown cartilage	27 (55.1)
Stage C2: Total coverage with white even smooth cartilage	10 (20.4)

Values are presented as no. (%)

[‡] Cartilage Regeneration Assessment of International Cartilage Repair Society

ICRS International Cartilage Repair Society

Discussion

The main finding of this study was that allogeneic hUCB-MSC implantation in patients undergoing HTO was safe and showed signs of cartilage status improvement for full-thickness cartilage defects in osteoarthritic knees. Moreover, it also provided excellent improvement of functional outcomes. When clinical outcomes were compared to cartilage regeneration status, the ICRS-CRA system showed no

significant correlation with clinical outcomes (KSS function score); hence, a larger study with a longer follow-up period will be required for more comprehensive analysis. Based on this analysis, our study may draw further conclusion whether improved cartilage regeneration produces better clinical outcomes in patients undergoing HTO.

Although HTO for correction of varus malalignment in the osteoarthritic knee provides good functional recovery and pain relief in the short-term [3, 20], cartilage regeneration is theoretically essential, because the articular surface may continue to degrade, leading to degenerative arthritis. Hence, the overall long-term success of HTO remains questionable without cartilage regeneration. Until now, the value of additional cartilage repair procedures during HTO for OA has been debated [21]. Among these procedures, microfracture surgery for cartilage repair is quite popular, but outcomes in osteoarthritic knees have been debated in many studies. Therefore, cell-based cartilage repair procedures using BMAC and adipose tissue-derived stromal vascular fraction (AD-SVF) cells have garnered great attention, even for use in osteoarthritic knees. Recently, several studies reported that clinical outcomes were significantly improved after HTO when BMAC or AD-SVF cells were applied [22, 23]. However, these procedures are relatively invasive [24, 25], and the percentage of mesenchymal stem cells (MSCs) in BMAC or AD-SVF from human patients is less than 5% [26]. Theoretically, single-staged procedures using allogeneic MSCs for cartilage regeneration would be more ideal than two-stage procedures using autologous MSCs. However, clinical use of allogeneic stem cells for cartilage regeneration has rarely been reported, and information on efficacy and safety is lacking. Recently,

Table 3 Clinical outcomes according to cartilage regeneration by ICRS and Koshino staging system*

ICRS grade				Koshino staging system		
	Good regeneration (n = 38)	Poor regeneration (n = 11)	p value [†]	Stage B (n = 12)	Stage C (n = 37)	p value [†]
IKDC	72.0 ± 6.2 (58–86)	72.0 ± 5.2 (64–78)	0.501	71.8 ± 5.6 (64–78)	71.9 ± 6.1 (58–86)	0.916
WOMAC						
Pain	2.7 ± 1.5 (0–6)	2.8 ± 1.1 (1–4)	0.447	2.4 ± 1.2 (1–4)	2.6 ± 1.4 (0–6)	0.615
Stiffness	2.0 ± 1.4 (0–5)	2.8 ± 1.9 (1–6)	0.716	2.7 ± 2.0 (1–6)	2.0 ± 1.4 (0–5)	0.395
Function	6.5 ± 2.8 (1–13)	6.8 ± 2.7 (3–13)	0.852	7.1 ± 2.8 (3–13)	6.3 ± 2.8 (1–13)	0.852
Total	11.2 ± 3.9 (1–24)	11.9 ± 4.0 (5–23)	0.566	12.2 ± 4.0 (5–23)	11.0 ± 3.8 (1–24)	0.454
KSS						
Pain	43.1 ± 5.7 (20–50)	44.0 ± 3.8 (40–50)	0.509	44.1 ± 3.6 (45–50)	43.0 ± 5.8 (20–50)	0.683
Function	84.1 ± 12.0 (50–100)	74.5 ± 11.2 (60–100)	0.059	76.6 ± 13.0 (60–100)	83.7 ± 11.9 (50–100)	0.100
HSS	85.6 ± 13.2 (31–100)	80.4 ± 13.9 (43–91)	0.101	81.3 ± 13.6 (43–91)	85.5 ± 13.4 (58–100)	0.217

*Values are presented as mean with standard deviation in parentheses. Italicized table entry indicates a p value of < 0.10 (marginally significant)

[†] The Mann-Whitney test was used to analyze differences, and a p value of < 0.05 was considered significant

ICRS International Cartilage Repair Society, IKDC International Knee Documentation Committee, WOMAC Western Ontario and McMaster Universities Osteoarthritis Index, KSS Knee Society Score, HSS Hospital for Special Surgery

application of commercial allogeneic hUCB-MSC products has drawn major interest, and there have been a few reports of actual stem cell therapy using autologous stem cells for the regenerative treatment of osteoarthritic cartilage defects [18, 27, 28].

A seven year follow-up of hUCB-MSC implantation into osteoarthritic knees revealed no osteogenesis or tumorigenesis, suggesting that this novel stem cell-based therapy is safe and effective in regeneration of durable articular cartilage in osteoarthritic knees [18]. Song et al. reported significant improvement in clinical outcomes of patients who underwent HTO along with hUCB-MSC implantation after a two year follow-up [13, 29]. Gobbi et al. reported successful long-term follow-up results of HA-BMAC implantation and that median clinical outcome scores over a short-term follow-up of two years were significantly increased from preoperative scores and not significantly different from scores over a long-term follow-up [25]. Similarly to their results, the present study shows that allogeneic hUCB-MSC implantation in patients undergoing HTO leads to excellent clinical outcomes.

The reliability of cartilage regenerative procedures remains controversial. Some studies have questioned the benefit of augmentation to HTO and demonstrated that the unloading effects of HTO provided improvement in both clinical outcomes and cartilage status [30, 31]. In a case series including second-look arthroscopy for cartilage regeneration in the medial compartment, Jung et al. reported that HTO without cartilage regenerative procedures resulted in regeneration of fibrocartilage in lesions of the MFC in 92% of knees [32]. However, only 4% of knees showed complete regeneration of fibrocartilage on the MFC. Goshima et al. evaluated changes in bone tracer uptake after HTO and demonstrated that HTO decreased bone tracer uptake regardless of cartilage regeneration and was associated with improved clinical outcomes [32]. On the other hand, several authors have previously suggested the application of autologous MSCs to provide superior cartilage regeneration accompanied by more favourable clinical outcomes in patients undergoing HTO [10, 23, 29, 33, 34]. In general, the repaired cartilage in such cases is phenotypically unstable cartilaginous tissue with inferior biochemical and biomechanical properties compared to those of native tissue [20]. In the present study, all 49 patients who received second-look arthroscopy showed good regeneration to normal or nearly normal grade, and only 10% of patients showed grade III regeneration. These results are similar to those of a previous study using hUCB-MSCs [29]. However, the present results seem similar to or better than those of other studies using BMAC or AD-VSF cells, although direct comparison is impossible due to different cell sources and procedures employed [35–37]. Recently, Koh et al. reported that 63% of

patients who were undergoing HTO and received microfracture surgery with adipose-derived MSC implantation were scored ICRS grade I or II, while only 55.5% of those receiving microfracture only were scored grade I or II [38]; these are slightly worse than the present results. However, because of the lack of a control group (HTO only) in the present study, the effect of hUCB-MSC itself remains unclear.

Although a few studies have shown better clinical outcomes when patients had good cartilage regeneration [6, 39], most have shown no such correlation [32, 40–42]. In the present study, cartilage regeneration according to ICRS-CRA score showed no significant correlation. A larger study with a longer follow-up period would reveal whether implantation of MSCs into patients undergoing HTO leads to greater cartilage remodeling and better clinical outcomes.

This study has some limitations to consider. First, this retrospective study lacked a control group receiving HTO alone to demonstrate the superiority of articular cartilage regeneration after cartilage regenerative procedures. Second, the follow-up period was relatively short, and the study included too few data points to draw solid conclusions. Nonetheless, hUCB-MSCs are a rather new medical product, given that no similar studies of this size have been published. Third, only those patients who wanted their metal plates removed received second-look arthroscopy. Finally, MRI or biopsy with histological evaluation would be a more reliable means of evaluating the properties of regenerated cartilage, but neither could be conducted solely for research purposes because of ethical issues related to possible morbidity and cost. Furthermore, the ICRS grades at second-look arthroscopic surgery were assessed by the surgeon, though independent assessment by three orthopaedic surgeons would have been preferable to eliminate bias. However, even independent judgments made in this way could only be made according to photographs taken during surgery, possibly introducing another bias. In addition, a power analysis with inter-rater variability of the articular cartilage grading system would be useful in future studies.

Conclusion

Allogeneic hUCB-MSC implantation concurrent with HTO was safe and showed signs of cartilage status improvement in HTO patients with symptomatic medial knee OA without any significant adverse effects. Thus, allogeneic hUCB-MSC implantation for enhanced cartilage regeneration may lead to better clinical outcomes in patients undergoing HTO. Although still in the early stages of application, MSC implantation seems to be useful for repairing cartilage lesions in osteoarthritic knees. Furthermore, randomized controlled studies with

a control group are necessary to determine the real effectiveness and indications of this new combined procedure for patients with osteoarthritis.

Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethics approval IRB approval was obtained prior to the start of this study.

Informed consent For this type of study, formal consent is not required.

References

- Rodriguez-Merchan EC (2016) Unicompartmental knee osteoarthritis (UKOA): unicompartmental knee arthroplasty (UKA) or high tibial osteotomy (HTO)? *Arch Bone Jt Surg* 4:307–313
- Cao Z, Mai X, Wang J, Feng E, Huang Y (2018) Unicompartmental knee arthroplasty vs high tibial osteotomy for knee osteoarthritis: a systematic review and meta-analysis. *J Arthroplast* 33:952–959. <https://doi.org/10.1016/j.arth.2017.10.025>
- Jin C, Song EK, Santoso A, Ingale PS, Choi IS, Seon JK (2020) Survival and risk factor analysis of medial open wedge high tibial osteotomy for unicompartment knee osteoarthritis. *Arthroscopy* 36:535–543. <https://doi.org/10.1016/j.arthro.2019.08.040>
- Wulfften Palthe AFY, Clement ND, Temmerman OPP, Burger BJ (2018) Survival and functional outcome of high tibial osteotomy for medial knee osteoarthritis: a 10–20-year cohort study. *Eur J Orthop Surg Traumatol* 28:1381–1389. <https://doi.org/10.1007/s00590-018-2199-6>
- Kanamiya T, Naito M, Hara M, Yoshimura I (2002) The influences of biomechanical factors on cartilage regeneration after high tibial osteotomy for knees with medial compartment osteoarthritis: clinical and arthroscopic observations. *Arthroscopy* 18:725–729. <https://doi.org/10.1053/jars.2002.35258>
- Koshino T, Wada S, Ara Y, Saito T (2003) Regeneration of degenerated articular cartilage after high tibial valgus osteotomy for medial compartmental osteoarthritis of the knee. *Knee* 10:229–236. [https://doi.org/10.1016/s0968-0160\(03\)00005-x](https://doi.org/10.1016/s0968-0160(03)00005-x)
- Kim MS, Koh JJ, Choi YJ, Pak KH, In Y (2017) Collagen augmentation improves the quality of cartilage repair after microfracture in patients undergoing high tibial osteotomy: a randomized controlled trial. *Am J Sports Med* 45:1845–1855. <https://doi.org/10.1177/0363546517691942>
- Yao RZ, Liu WQ, Sun LZ, Yu MD, Wang GL (2020) Effectiveness of high tibial osteotomy with or without other procedures for medial compartment osteoarthritis of knee: an update meta-analysis. *J Knee Surg* 138:227–236. <https://doi.org/10.1055/s-0039-1700978>
- Ding DC, Shyu WC, Lin SZ (2011) Mesenchymal stem cells. *Cell Transplant* 20:5–14. <https://doi.org/10.3727/096368910x>
- Koh YG, Jo SB, Kwon OR, Suh DS, Lee SW, Park SH et al (2013) Mesenchymal stem cell injections improve symptoms of knee osteoarthritis. *Arthroscopy* 29:748–755. <https://doi.org/10.1016/j.arthro.2012.11.017>
- Lopa S, Colombini A, Moretti M, de Girolamo L (2019) Injective mesenchymal stem cell-based treatments for knee osteoarthritis: from mechanisms of action to current clinical evidences. *Knee Surg Sports Traumatol Arthrosc* 27:2003–2020. <https://doi.org/10.1007/s00167-018-5118-9>
- Yang SE, Ha CW, Jung M, Jin HJ, Lee M, Song H et al (2004) Mesenchymal stem progenitor cells developed in cultures from UC blood. *Cytotherapy* 6:476–486. <https://doi.org/10.1080/14653240410005041>
- Song JS, Hong KT, Kim NM, Jung JY, Park HS, Lee SH et al (2020) Implantation of allogenic umbilical cord blood-derived mesenchymal stem cells improves knee osteoarthritis outcomes: two-year follow-up. *Regen Ther* 14:32–39. <https://doi.org/10.1016/j.reth.2019.10.003>
- Gao J, Dennis JE, Solchaga LA, Awadallah AS, Goldberg VM, Caplan AI (2001) Tissue-engineered fabrication of an osteochondral composite graft using rat bone marrow-derived mesenchymal stem cells. *Tissue Eng* 7:363–371. <https://doi.org/10.1089/10763270152436427>
- Floerkemeier S, Staubli AE, Schroeter S, Goldhahn S, Lobenhoffer P (2013) Outcome after high tibial open-wedge osteotomy: a retrospective evaluation of 533 patients. *Knee Surg Sports Traumatol Arthrosc* 21:170–180. <https://doi.org/10.1007/s00167-012-2087-2>
- Lee DK, Kim KK, Ham CU, Yun ST, Kim BK, Oh KJ (2018) The learning curve for biplane medial open wedge high tibial osteotomy in 100 consecutive cases assessed using the cumulative summation method. *Knee Surg Relat Res* 30:303–310. <https://doi.org/10.5792/ksrr.17.064>
- Fujisawa Y, Masuhara K, Shiomi S (1979) The effect of high tibial osteotomy on osteoarthritis of the knee. An arthroscopic study of 54 knee joints. *Orthop Clin North Am* 10:585–608
- Park YB, Ha CW, Lee CH, Yoon YC, Park YG (2017) Cartilage regeneration in osteoarthritic patients by a composite of allogeneic umbilical cord blood-derived mesenchymal stem cells and hyaluronate hydrogel: results from a clinical trial for safety and proof-of-concept with 7 years of extended follow-up. *Stem Cells Transl Med* 6:613–621. <https://doi.org/10.5966/sctm.2016-0157>
- Brittberg M, Winalski CS (2003) Evaluation of cartilage injuries and repair. *J Bone Joint Surg Am* 2:58–69. <https://doi.org/10.2106/00004623-200300002-00008>
- Ekeland A, Nerhus TK, Dimmen S, Thomsen E, Heir S (2017) Good functional results following high tibial opening-wedge osteotomy of knees with medial osteoarthritis: a prospective study with a mean of 8.3 years of follow-up. *Knee* 24:380–389. <https://doi.org/10.1016/j.knee.2016.12.005>
- Lee OS, Ahn S, Ahn JH, Teo SH, Lee YS (2018) Effectiveness of concurrent procedures during high tibial osteotomy for medial compartment osteoarthritis: a systematic review and meta-analysis. *Arch Orthop Trauma Surg* 138:227–236. <https://doi.org/10.1007/s00402-017-2826-4>
- Cavallo M, Sayyed-Hosseini SH, Parma A, Buda R, Mosca M, Giannini S (2018) Combination of high tibial osteotomy and autologous bone marrow derived cell implantation in early osteoarthritis of knee: a preliminary study. *Arch Bone Jt Surg* 6:112–118
- Kyriakidis T, Iosifidis M, Michalopoulos E, Melas I, Stavropoulos-Giokas C, Verdonk R (2020) Good mid-term outcomes after adipose-derived culture-expanded mesenchymal stem cells implantation in knee focal cartilage defects. *Knee Surg Sports Traumatol Arthrosc* 28:502–508. <https://doi.org/10.1007/s00167-019-05688-9>
- Cotter EJ, Wang KC, Yanke AB, Chubinskaya S (2018) Bone marrow aspirate concentrate for cartilage defects of the knee: from bench to bedside evidence. *Cartilage* 9:161–170. <https://doi.org/10.1177/1947603517741169>
- Gobbi A, Whyte GP (2019) Long-term clinical outcomes of one-stage cartilage repair in the knee with hyaluronic acid-based scaffold embedded with mesenchymal stem cells sourced from bone marrow aspirate concentrate. *Am J Sports Med* 47:1621–1628. <https://doi.org/10.1177/0363546519845362>
- Shapiro SA, Kazmerchak SE, Heckman MG, Zubair AC, O'Connor MI (2017) A prospective, single-blind, placebo-

- controlled trial of bone marrow aspirate concentrate for knee osteoarthritis. *Am J Sports Med* 45:82–90. <https://doi.org/10.1177/0363546516662455>
27. De Bari C, Roelofs AJ (2018) Stem cell-based therapeutic strategies for cartilage defects and osteoarthritis. *Curr Opin Pharmacol* 40:74–80. <https://doi.org/10.1016/j.coph.2018.03.009>
 28. Freitag J, Bates D, Boyd R, Shah K, Barnard A, Huguenin L et al (2016) Mesenchymal stem cell therapy in the treatment of osteoarthritis: reparative pathways, safety and efficacy - a review. *BMC Musculoskelet Disord* 17:230–242. <https://doi.org/10.1186/s12891-016-1085-9>
 29. Song JS, Hong KT, Kim NM, Park HS, Choi NH (2020) Human umbilical cord blood-derived mesenchymal stem cell implantation for osteoarthritis of the knee. *Arch Orthop Trauma Surg* 140:503–509. <https://doi.org/10.1007/s00402-020-03349-y>
 30. Genovese G, Tournaki A, Berti E, Brambilla L (2019) Comment on: ‘Treatment of psoriasis with ustekinumab in a patient with HIV-related Kaposi sarcoma’. *Clin Exp Dermatol* 44(4):e147–e148. <https://doi.org/10.1111/ced.13935>
 31. Filardo G, Zaffagnini S, De Filippis R, Perdida F, Andriolo L, Candrian C (2018) No evidence for combining cartilage treatment and knee osteotomy in osteoarthritic joints: a systematic literature review. *Knee Surg Sports Traumatol Arthrosc* 26:3290–3299. <https://doi.org/10.1007/s00167-018-4871-0>
 32. Jung WH, Takeuchi R, Chun CW et al (2014) Second-look arthroscopic assessment of cartilage regeneration after medial opening-wedge high tibial osteotomy. *Arthroscopy* 30:72–79. <https://doi.org/10.1016/j.arthro.2013.10.008>
 33. Ding DC, Chang YH, Shyu WC, Lin SZ (2015) Human umbilical cord mesenchymal stem cells: a new era for stem cell therapy. *Cell Transplant* 24:339–347. <https://doi.org/10.3727/096368915x686841>
 34. Zhang X, Hirai M, Cantero S, Ciubotariu R, Dobrila L, Hirsh A et al (2011) Isolation and characterization of mesenchymal stem cells from human umbilical cord blood: reevaluation of critical factors for successful isolation and high ability to proliferate and differentiate to chondrocytes as compared to mesenchymal stem cells from bone marrow and adipose tissue. *J Cell Biochem* 112:1206–1218. <https://doi.org/10.1002/jcb.23042>
 35. Buda R, Vannini F, Cavallo M, Grigolo B, Cenacchi A, Giannini S (2010) Osteochondral lesions of the knee: a new one-step repair technique with bone-marrow-derived cells. *J Bone Joint Surg Am* 2:2–11. <https://doi.org/10.2106/jbjs.J.00813>
 36. Gigante A, Calcagno S, Cecconi S, Ramazzotti D, Manzotti S, Enea D (2011) Use of collagen scaffold and autologous bone marrow concentrate as a one-step cartilage repair in the knee: histological results of second-look biopsies at 1 year follow-up. *Int J Immunopathol Pharmacol* 24:69–72. <https://doi.org/10.1177/03946320110241s213>
 37. Giannini S, Buda R, Vannini F, Cavallo M, Grigolo B (2009) One-step bone marrow-derived cell transplantation in talar osteochondral lesions. *Clin Orthop Relat Res* 467:3307–3320. <https://doi.org/10.1007/s11999-009-0885-8>
 38. Koh YG, Kwon OR, Kim YS, Choi YJ, Tak DH (2016) Adipose-derived mesenchymal stem cells with microfracture versus microfracture alone: 2-year follow-up of a prospective randomized trial. *Arthroscopy* 32:97–109. <https://doi.org/10.1016/j.arthro.2015.09.010>
 39. Koh YG, Kwon OR, Kim YS, Choi YJ (2014) Comparative outcomes of open-wedge high tibial osteotomy with platelet-rich plasma alone or in combination with mesenchymal stem cell treatment: a prospective study. *Arthroscopy* 30:1453–1460. <https://doi.org/10.1016/j.arthro.2014.05.036>
 40. Matsunaga D, Akizuki S, Takizawa T, Yamazaki I, Kuraishi J (2007) Repair of articular cartilage and clinical outcome after osteotomy with microfracture or abrasion arthroplasty for medial gonarthrosis. *Knee* 14:465–471. <https://doi.org/10.1016/j.knee.2007.06.008>
 41. Kim CW, Seo SS, Lee CR, Gwak HC, Kim JH, Jung SG (2017) Factors affecting articular cartilage repair after open-wedge high tibial osteotomy. *Knee* 24:1099–1107. <https://doi.org/10.1016/j.knee.2017.06.001>
 42. Odenbring S, Egund N, Lindstrand A, Lohmander LS, Willen H (1992) Cartilage regeneration after proximal tibial osteotomy for medial gonarthrosis. An arthroscopic, roentgenographic, and histologic study. *Clin Orthop Relat Res* 277:210–216

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.